

Report 2 - Minimal Design and Prototype

Project III - Tools for Program Development

System Architecture

DevTrack uses a client-server architecture.

React frontend -> Express REST API -> SQLite database

Frontend Design

The frontend is a single-page React application. It contains:

- Login screen
- Project sidebar
- Dashboard statistics
- Kanban issue board
- Issue filters and search
- Issue detail panel
- New issue and new project dialogs

Backend Design

The backend is an Express API. It exposes endpoints for authentication, users, projects, issues, comments, and dashboard statistics.

Database Design

Main tables:

- `users`
- `projects`
- `issues`
- `comments`
- `activity`

Prototype Status

The prototype includes:

- Working login
- Seeded users, projects, issues, comments, and activity
- Project list
- Issue creation
- Issue status updates
- Issue detail view
- Comments
- Dashboard counters
- Filtering and search

API Overview

Method	Endpoint	Description
POST	<code>/api/auth/login</code>	Authenticate user
GET	<code>/api/users</code>	List team members
GET	<code>/api/projects</code>	List projects
POST	<code>/api/projects</code>	Create project
GET	<code>/api/issues</code>	List and filter issues
POST	<code>/api/issues</code>	Create issue
GET	<code>/api/issues/:id</code>	View issue details
PATCH	<code>/api/issues/:id</code>	Update issue
POST	<code>/api/issues/:id/comments</code>	Add comment
GET	<code>/api/dashboard/stats</code>	View dashboard data